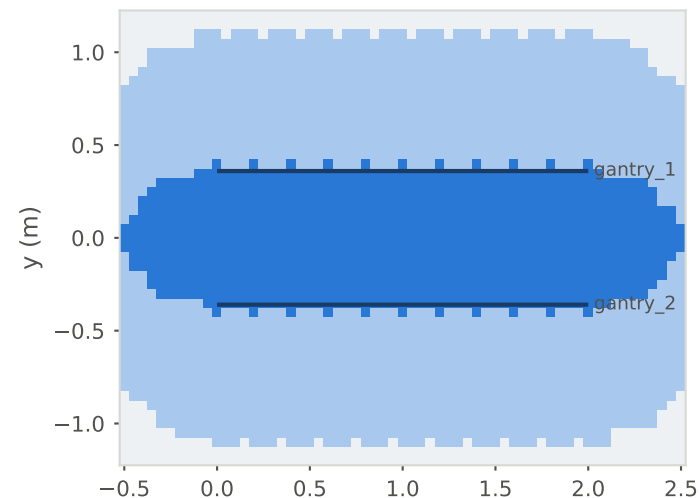
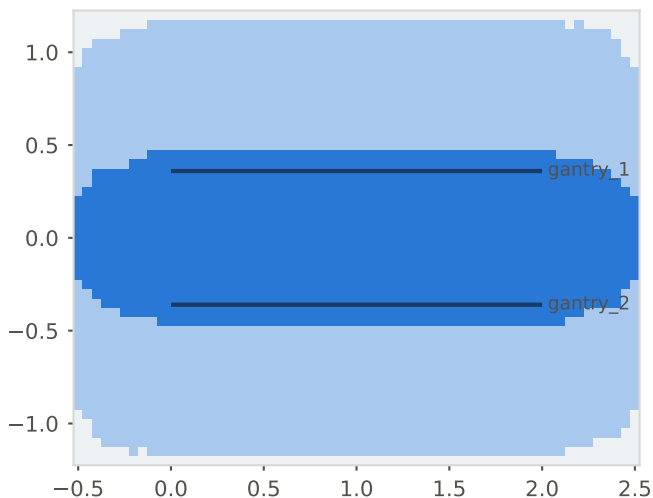


Co-manipulation capability map — 0.25 m block, top-face corners, 4 arms, 2 shared gantries

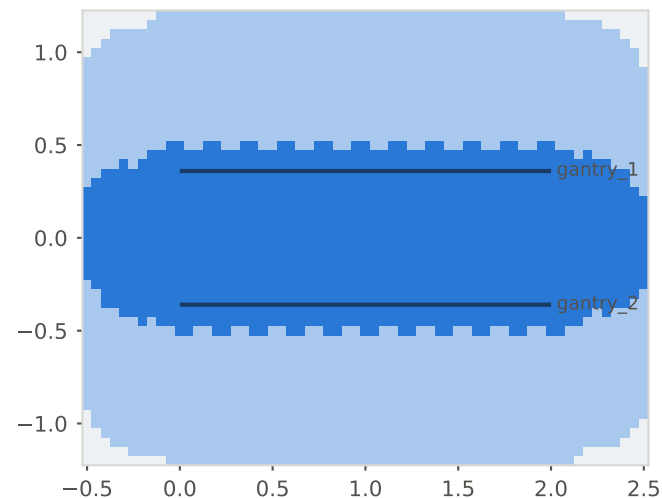
$z = 1.2 \text{ m}$, $\text{yaw} = 0^\circ$
28.6% feasible



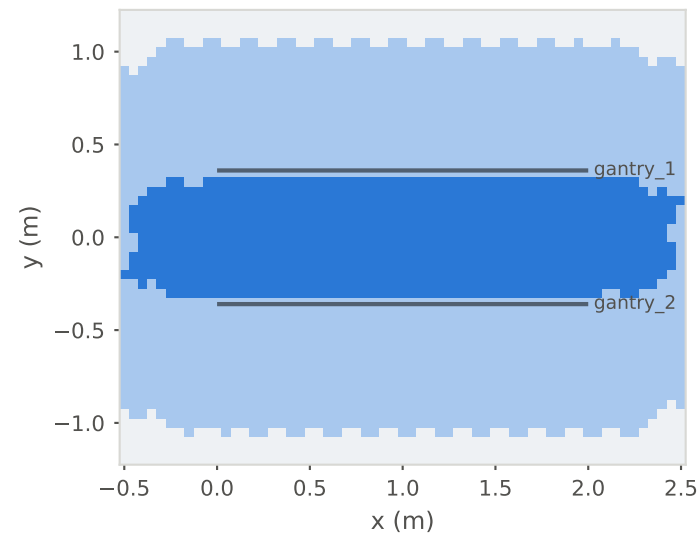
$z = 1.4 \text{ m}$, $\text{yaw} = 0^\circ$
36.2% feasible



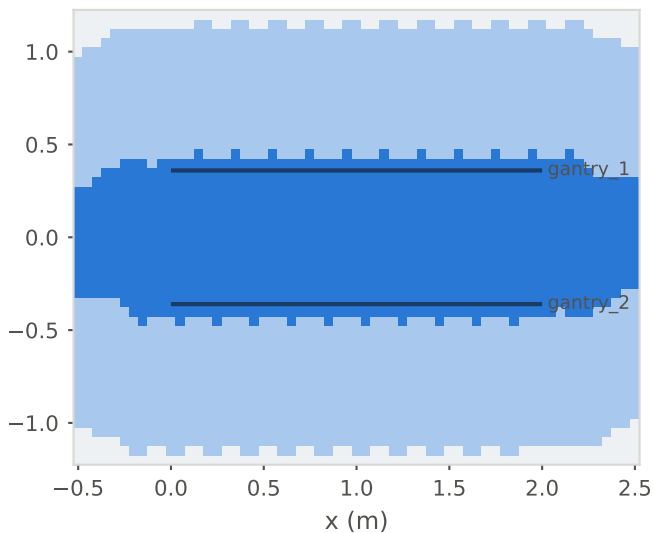
$z = 1.6 \text{ m}$, $\text{yaw} = 0^\circ$
38.2% feasible



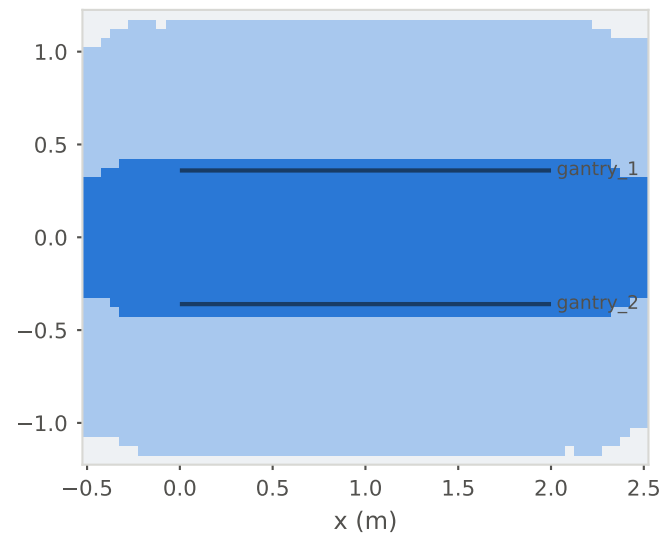
$z = 1.2 \text{ m}$, $\text{yaw} = 45^\circ$
24.6% feasible



$z = 1.4 \text{ m}$, $\text{yaw} = 45^\circ$
34.0% feasible



$z = 1.6 \text{ m}$, $\text{yaw} = 45^\circ$
33.8% feasible



no gantry pair one gantry pair only all four arms